

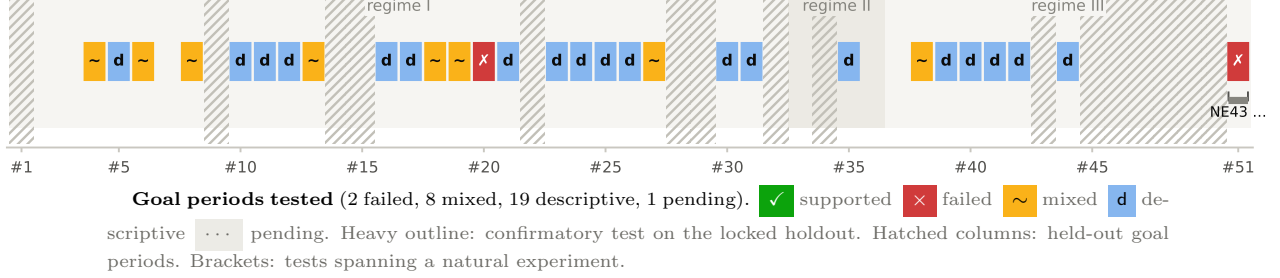
## H20: Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

**Bottom line.** LLM-swarm content shows no detectable aging at day resolution: after a few days' kickoff relaxation it decorrelates at an age-independent rate, and only a null calibrated to the low-dimensional fluctuation shape says so honestly.

level: hypothesis · pipeline checks 100% · updated 2026-10-03



### WHAT HAPPENED

We measured the two-time correlation  $C(t_w + \tau, t_w)$  of each agent's day-mean content embedding ( $t_w$  = active days since kickoff) in 29 non-holdout goal periods. The aging slope  $A$  at fixed lag ( $t_w \geq 2$ ) was tested against a fitted stationary swarm model. Synthetic check at village sampling: size 0.03–0.07, power 0.33–0.44 against  $\mu = 0.5$  aging. The pre-registered isotropic null was 2–3× too narrow on real data (content fluctuates in 5–12 of 32 dimensions); a post-hoc null with the fitted shape fixed it. **No aging:** random-effects  $A$  over long and medium goals is  $+0.021 \pm 0.021$ . The one strong effect (#38) is a kickoff relaxation: lag-1  $C$  rises 0.66→0.87 in four days, then stays flat. In #51,  $\mu = 0.5$  aging is rejected. Holidays drive the short-week outliers. Holdout not run.

### PHYSICAL MODEL

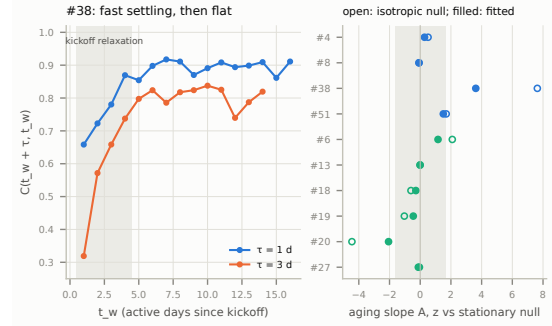
Models 01 (SK-type glassy aging) and 11 (vector spins). Spin: agent  $i$ 's day state  $\bar{x}_{it}$ , the mean of its unit-normalized, regime-whitened ( $n = 32$ ) statement embeddings (own chat and plans);  $v_{it}$  is its sampling noise.  $q$ : plateau (style and goal content);  $c_0$ : short-lag ceiling;  $\tau_0$ : relaxation time;  $\mu$ : aging exponent of the Box-Cox clock  $s_\mu$  ( $\mu = 0$  stationary,  $\mu = 1$  simple aging). Rival: a kickoff transient decaying in  $\tau_q$  days.  $n_{\text{eff}}$ : dimension of the day-to-day fluctuations.

$$C(t, t') = \frac{\sum_i \bar{x}_{it} \cdot \bar{x}_{it'}}{\sqrt{\sum_i S_{it} \sum_i S_{it'}}}, \quad S_{it} = |\bar{x}_{it}|^2 - v_{it}$$

$$C(t_w + \tau, t_w) = q + (c_0 - q) e^{-\Delta s / \tau_0}$$

$$\Delta s = s_\mu(t_w + \tau) - s_\mu(t_w), \quad s_\mu(t) = \frac{t^{1-\mu}-1}{1-\mu}$$

$$A = \partial C / \partial \ln t_w \big|_\tau, \quad \text{Var } C(t, t') \propto 1/n_{\text{eff}}$$



**Observables.** Left: in #38, the agent-averaged correlation between an agent's content on day  $t_w$  and on day  $t_w + \tau$  climbs for four active days after the kickoff, then stays flat. A quench relaxes; the system does not slow with age. Right: the aging slope  $A$  as a z-score against the stationary null, for long (blue) and medium (aqua) goals. Open circles: the pre-registered isotropic null (too narrow). Filled: the null with the fitted fluctuation shape. Only #38 (up) and #20 (Thanksgiving, down) stand out.

### SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a negative result, mildly useful. An operator worried that a long-running swarm "rigidifies" as its goal ages can relax a little: at day resolution, content turnover does not slow in week three versus week one (weak aging is not excluded). The practical number is the settling time after a big kickoff, about four active days, seen once (#38). The more transferable lesson is methodological: embedding similarity statistics on swarm logs need nulls with the empirical, low-dimensional fluctuation shape, or they are 2–3× overconfident. As physics, aging finds nothing to grip: daily re-prompting with a fixed goal acts like a strong pinning field.

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## RESULTS VS PREDICTIONS

*Pending.*

### GOAL PERIODS IN DETAIL

G04 ~ mixed (underpowered: not significant,  $\mu = 0.5$  aging not rejected) (...  
 G05 d descriptive (stationary contrast) (Amendment-2 null; pre-registered...  
 G06 ~ mixed (underpowered: not significant,  $\mu = 0.5$  aging not rejected) (...  
 G08 ~ mixed (underpowered: not significant,  $\mu = 0.5$  aging not rejected) (...  
 G10 d descriptive (stationary contrast) (Amendment-2 null; pre-registered...  
 G11 d descriptive (stationary contrast) (Amendment-2 null; pre-registered...  
 G12 d descriptive (stationary contrast) (Amendment-2 null; pre-registered...  
 G13 ~ mixed (underpowered: not significant,  $\mu = 0.5$  aging not rejected) (...  
 G16 d descriptive (stationary contrast) (Amendment-2 null; pre-registered...  
 G17 d descriptive (stationary contrast) (Amendment-2 null; pre-registered...  
 G18 ~ mixed (underpowered: not significant,  $\mu = 0.5$  aging not rejected) (...  
 G19 ~ mixed (underpowered: not significant,  $\mu = 0.5$  aging not rejected) (...  
 G20  $\times$  failed (dynamics speed up with age,  $A_j > 0$ ) (Amendment-2 null; pre-r...  
 G21 d descriptive (stationary contrast) (Amendment-2 null; pre-registered...  
*+16 more periods: see the card.*

### FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 0 |
| D unfitted predictions | 0 | I transfer        | 0 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

### CAVEATS

*Pending.*

### ROUND 2

- **Essence.** The swarm has no long-term internal memory; all persistence is stored outside the agents.
- **H20-R1** Persistent content order is explained entirely by persistent fields (goal prompt, artifacts); without them content decorrelates within a day (E3, E4).
- **H20-R2** The ~2–4 day kickoff relaxation is set by artifact build-up, not agent memory.

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